CRIME TREND ANALYSIS AND FORECASTING IN PAKISTAN

A Statistical and Machine Learning Approach Using ARIMA and ETS Models

Independent Data Analysis Project

By

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ABOUT THIS PROJECT

This report presents an independent data analysis project undertaken by Manzoor Ali and Umer Riaz, applying statistical and machine learning methods to publicly available, national-level crime data for Pakistan. It was completed outside any academic program or institutional requirement, as a self-directed exercise in applied data analysis, and is shared here as a portfolio piece.

Data source: Ministry of Interior, National Police Bureau, Islamabad.

TOOLS & SKILLS DEMONSTRATED

- Programming & data wrangling: R, tidyverse (dplyr, tidyr), readxl
- Data cleaning: anomaly detection, structural-break identification, imputation
- Data visualization: ggplot2 (trend, composition, seasonality, distribution charts)
- Statistical inference: Shapiro-Wilk normality testing, one-way ANOVA, Kruskal-Wallis test
- Regression modelling: linear trend and trend + seasonal models
- Unsupervised machine learning: k-means clustering, PCA visualization
- Time-series analysis: classical decomposition, ACF, stationarity testing (ADF, KPSS)
- Forecasting: ARIMA (auto.arima), ETS exponential smoothing, forecast accuracy comparison (RMSE, MAE, MAPE)

ABSTRACT

Crime is a recurring concern for law enforcement and policymakers, yet quantitative, data-driven analysis of national crime trends remains limited in Pakistan. This project presents an end-to-end statistical and machine learning analysis of month-wise crime data reported across Pakistan between January 2022 and December 2024, sourced from the Ministry of Interior's National Police Bureau. The dataset covers eleven crime categories, including nine core categories reported consistently across all three years (Murder, Attempted Murder, Kidnapping/Abduction, Dacoity, Robbery, Burglary, Cattle Theft, Other Theft, and Miscellaneous), with Rape and Gang Rape reported separately from 2024 onward.

The analysis follows a structured pipeline comprising data import, cleaning, and transformation; exploratory data analysis; statistical hypothesis testing (Shapiro-Wilk, one-way ANOVA, Kruskal-Wallis); regression modelling; k-means clustering; time-series decomposition; and forecasting using ARIMA and ETS models, implemented in R.

Results show that total reported crime rose from approximately 1.12 million incidents in 2022 to 1.51 million in 2024, a 35% increase over the three-year window, with Robbery and Burglary growing fastest among core categories (81% and 71% respectively) and Dacoity declining slightly (-2%). A linear time trend explained 57% of month-to-month variation in total crime, rising to 67% once calendar-month seasonal effects were added. Total crime displays a consistent seasonal pattern, peaking around March and August-September and troughing in January, April, June, and July, though a Kruskal-Wallis test on this pattern did not reach statistical significance given the short series. K-means clustering separated high-volume categories (Miscellaneous, Other Theft) from smaller violent-crime categories that move together. Both ARIMA and ETS models, fitted to the total monthly crime series, project continued growth through 2025, with projected monthly totals broadly in the range of 100,000 to 158,000 incidents and the seasonal pattern preserved.

The project concludes that reported crime in Pakistan has followed a clear upward trajectory over 2022-2024 with an identifiable seasonal component, and that combined statistical and machine learning methods provide a practical, replicable framework for monitoring and forecasting crime trends. Key limitations include the short 36-month series, an apparent reporting or classification change affecting the Robbery series from February 2023 onward, and reliance on reported rather than actual crime incidence.

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CHAPTER 1:
INTRODUCTION

1.1 Background

Monitoring how crime evolves over time is a basic requirement for effective policing, resource planning, and public policy. In Pakistan, monthly crime statistics are compiled by the Ministry of Interior through the National Police Bureau, covering multiple categories of reported crime across the country. While these records are collected routinely, they are not always subjected to systematic statistical or predictive analysis capable of answering practical questions such as: Is crime rising or falling? Are some categories growing faster than others? Does crime follow a seasonal pattern that could inform the timing of policing resources? What might the next twelve months look like?

This project applies both classical statistical methods (descriptive statistics, hypothesis testing, regression) and machine learning techniques (k-means clustering, ARIMA and ETS time-series forecasting) to month-wise national crime data for Pakistan covering January 2022 to December 2024, within a single, reproducible analytical pipeline implemented in R.

1.2 Problem Statement

Despite the routine collection of national crime statistics in Pakistan, there is a lack of publicly documented, integrated quantitative analysis that combines trend estimation, seasonality detection, statistical testing, and short-term forecasting for reported crime. This project addresses that gap by applying a structured statistical and machine learning framework to three years of national crime data.

1.3 Objectives

- To examine the overall trend in total reported crime in Pakistan between January 2022 and December 2024.
- To identify seasonal patterns and the category-level composition of reported crime.
- To statistically test whether differences across crime categories and calendar months are significant.

- To model the relationship between time and total crime using regression, with and without seasonal effects.
- To cluster crime categories and months by similarity in their temporal behaviour.
- To forecast total monthly crime for 2025 using ARIMA and ETS models and compare their performance.

1.4 Research Questions

- RQ1: How has total reported crime in Pakistan changed between 2022 and 2024?
- RQ2: Which crime categories have grown fastest, and which have declined, over the study period?
- RQ3: Is there a statistically identifiable seasonal pattern in total reported crime?
- RQ4: How much of the variation in total crime can be explained by a simple time trend, and does adding seasonality improve this?
- RQ5: Do crime categories and months form meaningful clusters based on their temporal behaviour?
- RQ6: What do ARIMA and ETS models project for total crime in 2025, and how do the two approaches compare?

1.5 Scope and Limitations

The project is limited to national-level, monthly crime totals across eleven reported categories for the period January 2022 to December 2024 (36 months); it does not analyze province- or district-level data, nor does it attempt to establish the socioeconomic or policy causes behind observed trends. Forecasts are limited to a twelve-month horizon (2025). Three limitations should be borne in mind throughout: the series is short (36 monthly observations), which constrains the precision of seasonal and forecasting models; the Robbery category shows an apparent reporting or classification change from February 2023 onward that should be treated as a structural break rather than a genuine spike; and one data point (Other Theft, April 2022) was identified as a likely transcription error and imputed for analysis. These issues are documented in Chapter 3 and revisited in Chapter 5.

1.6 Structure of This Report

Chapter 2 gives brief background on the methods used and why they suit this kind of data. Chapter 3 describes the data source and the methodology used for data import, cleaning, transformation, and analysis, including the software and evaluation criteria applied. Chapter 4 presents the results of the exploratory, statistical, regression, clustering, time-series, and forecasting analyses, together with a discussion of the findings. Chapter 5 summarizes conclusions, limitations, and recommendations.

CHAPTER 2:
BACKGROUND AND METHODS CONTEXT

Quantitative, data-driven analysis of national crime trends is comparatively limited in Pakistan, with much existing public discussion of crime relying on descriptive reporting of police statistics rather than integrated statistical or forecasting frameworks. Internationally, national crime records — such as the Uniform Crime Reporting (UCR) program in the United States and Home Office statistics in the United Kingdom — have long served as the basis for more systematic trend and forecasting analysis, typically combining descriptive statistics with formal hypothesis testing and, increasingly, time-series and machine learning methods.

The methods applied in this project are standard, well-established tools chosen to fit monthly count data with trend and seasonal components. Descriptive statistics and normality testing (Shapiro-Wilk) establish the basic shape of each category's data and whether parametric methods are appropriate; ANOVA and the Kruskal-Wallis test assess whether differences between categories or calendar months are statistically meaningful; linear regression quantifies the time trend and the incremental value of adding seasonality; k-means clustering groups categories or months by similarity in behaviour rather than raw scale; and ARIMA (Box & Jenkins, 1970) and ETS (Hyndman & Athanasopoulos, 2021) are two of the most widely used approaches for univariate time-series forecasting, well suited to this kind of data and commonly compared side-by-side using standard accuracy metrics (RMSE, MAE, MAPE).

This project brings these methods together into a single, reproducible pipeline — applied specifically to Pakistani national crime data — with explicit attention to data-quality issues (structural breaks, anomalies) that are often overlooked in purely descriptive treatments of crime statistics.

CHAPTER 3:
METHODOLOGY

3.1 Data Source and Description

The data used in this study is month-wise national crime data for Pakistan, sourced from the Ministry of Interior's National Police Bureau, Islamabad, covering January 2022 to December 2024. The dataset records eleven crime categories: Murder, Attempted Murder, Kidnapping/Abduction, Dacoity, Robbery, Burglary, Cattle Theft, Other Theft, Miscellaneous, Rape, and Gang Rape. Of these, nine "core" categories (all except Rape and Gang Rape) are reported consistently across all three years; Rape and Gang Rape are reported as separate categories only from 2024 onward and were therefore excluded from year-over-year comparisons but retained in the 2024 category-share analysis.

3.2 Data Import Procedure

The raw dataset was supplied as a single worksheet (Moth-wise-Crime-data.xlsx) containing three stacked year-blocks (2022, 2023, 2024), each consisting of a year header, a Year/Month column header, one row per crime category, and monthly counts across twelve columns (January-December) plus a yearly total column. Because the blocks are stacked rather than presented in a single tabular structure, a custom import routine was written in R to scan down the sheet row by row, track the current year whenever a four-digit year marker was encountered, and read each subsequent crime-category row into a normalized long-format table with columns Year, Month, CrimeType, and Count. This procedure produced 348 category-month observations (11 categories × 12 months across 3 years, with Rape and Gang Rape present only in 2024).

3.3 Data Cleaning Procedure

3.3.1 Structural Checks

The imported long-format table was checked for missing cells and duplicate Year-Month-CrimeType combinations; none were found.

3.3.2 Anomaly Detection and Correction

One data-entry anomaly was identified: Other Theft for April 2022 was recorded as 900, far below the neighbouring months for that category (March 2022: 12,818; May 2022: 11,596) and well outside the category's typical monthly range of roughly 11,000-28,000 across the full series. This value was treated as a probable transcription error, retained in the dataset as a separate Count_raw field for transparency, and replaced for downstream analysis with the average of the March and May 2022 values (12,207).

3.3.3 Structural Break — Robbery Series

Robbery counts rise sharply from 2,564 in January 2023 to 8,227 in February 2023 and remain elevated (roughly 5,500-8,600 per month) for the remainder of the series, compared with a 2022 range of about 2,900-4,250. Because this step-change persists for nearly two years rather than reverting, it is more consistent with a change in reporting scope or classification than with a genuine, sustained crime spike. In the absence of an official source confirming such a change, the values were retained as reported, but are treated throughout this report as a structural break that should be considered when interpreting Robbery trend and forecast results.

3.3.4 Category Naming

Category labels were standardized for consistent capitalization and spacing (for example, “Cattle theft” was standardized to “Cattle Theft” and “Other theft” to “Other Theft”).

3.4 Data Transformation Procedure

The cleaned long-format table was extended with a Date field (the first day of each month) and used to construct three derived structures: a monthly TOTAL series (the sum of all reported categories per month), used as the primary series for time-series analysis and forecasting; a wide-format table with crime categories as columns, used for correlation analysis and clustering; and year-over-year category totals with percentage change from 2022 to 2024, used for exploratory growth charts. The nine core categories present in all three years were used for all cross-year comparisons.

3.5 Analytical Framework

The analysis followed a sequential framework: exploratory data analysis (trend, composition, seasonality, and category relationships) established the descriptive picture; formal statistical tests (normality, ANOVA, Kruskal-Wallis) assessed whether observed differences were statistically meaningful; regression analysis quantified the time trend and the incremental contribution of seasonality; k-means clustering grouped categories and months by behavioural similarity; classical time-series decomposition separated trend, seasonal, and residual components of total crime; and ARIMA and ETS models were fitted to produce twelve-month-ahead forecasts for 2025, compared using standard accuracy metrics.

3.6 Software and Tools

All analysis was conducted in R. Data import and wrangling used base R and the tidyverse packages (readxl for reading the source spreadsheet, dplyr and tidyr for reshaping and cleaning); visualization used ggplot2; time-series decomposition, ARIMA, and ETS modelling used the forecast and stats packages (auto.arima() and ets()); stationarity checks used the tseries package (Augmented Dickey-Fuller and KPSS tests); and clustering and its visualization used the stats and factoextra packages (k-means and PCA projection). [Note: confirm the exact package list and R version used against your own script before final submission.]

3.7 Model Evaluation Criteria

Forecast accuracy for the ETS and ARIMA models was assessed using standard error metrics: Root Mean Squared Error (RMSE), which penalizes larger errors more heavily; Mean Absolute Error (MAE), which reflects the average magnitude of forecast error in the original units (number of crimes); and Mean Absolute Percentage Error (MAPE), which expresses error as a percentage of actual values, allowing comparison across series of different scale. Model selection for the linear trend and trend-plus-seasonal regressions used the coefficient of determination (R^2), which measures the proportion of variance in total crime explained by each model. For ARIMA model order selection, auto.arima() used the corrected Akaike Information Criterion (AICc), which balances model fit against complexity to avoid overfitting.

CHAPTER 4:
RESULTS AND DISCUSSION

4.1 Exploratory Data Analysis

Twenty-four figures were produced covering trend, seasonality, distribution, composition, and category relationships across the three-year study period. Key figures are presented in this section.

4.1.1 Overall Trend

Figure 1. *Total monthly crimes in Pakistan, Jan 2022 - Dec 2024.*

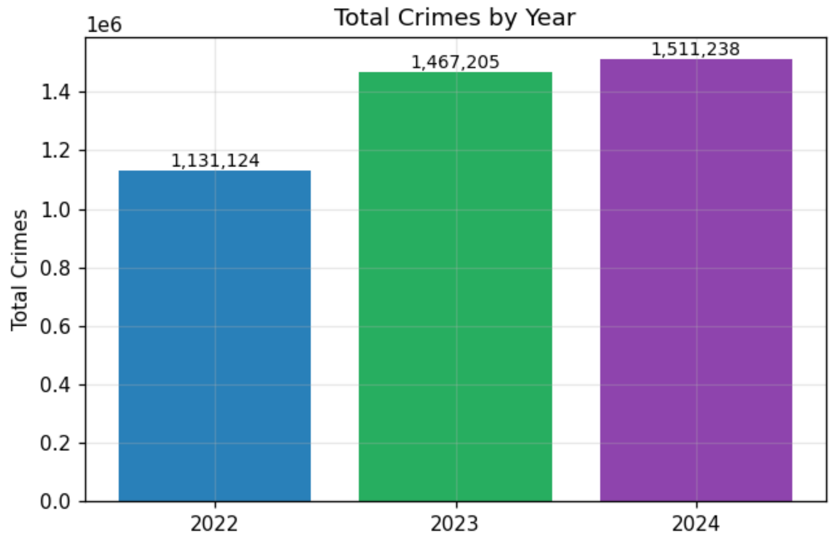

Figure 2. *Total reported crimes by year: 1,119,817 (2022); 1,467,205 (2023); 1,511,238 (2024).*

Total reported crime rose steadily across the three years, with the largest year-on-year increase occurring between 2022 and 2023.

4.1.2 Category Composition

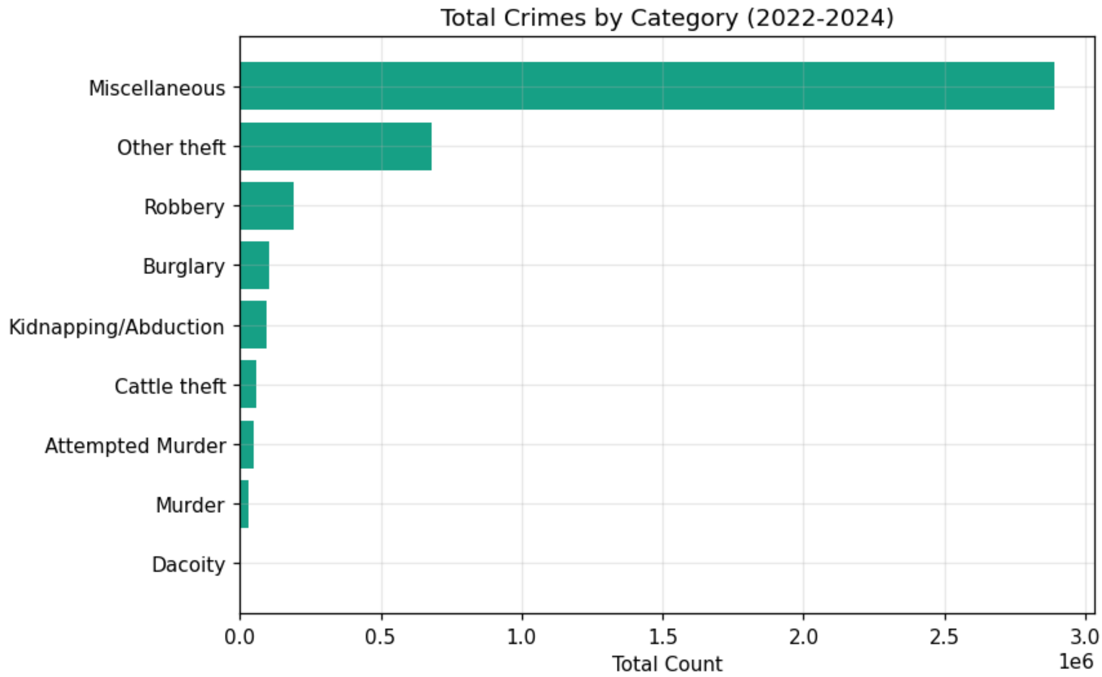

Figure 3. *Total crimes by category, summed across 2022-2024.*

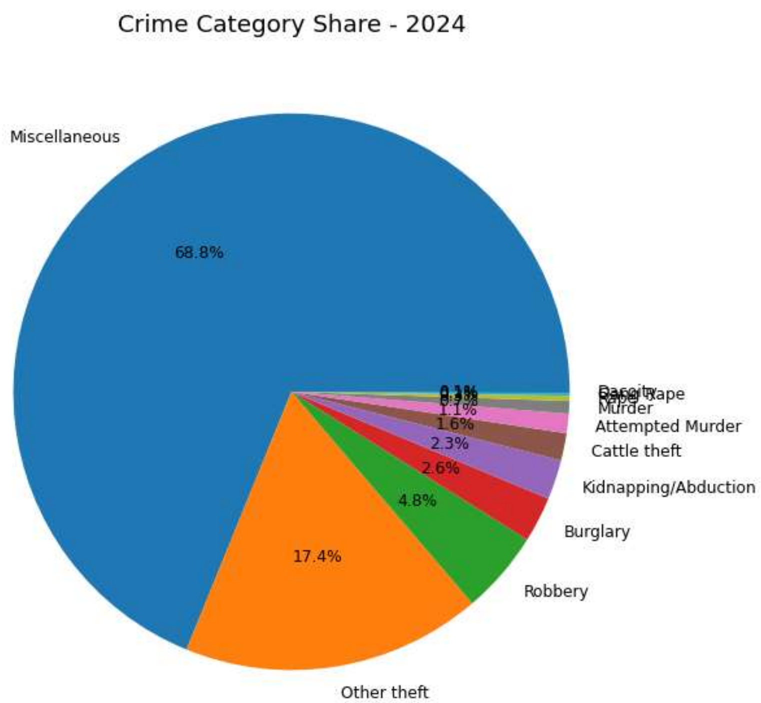

Figure 4. *Category share of total crime, 2024.*

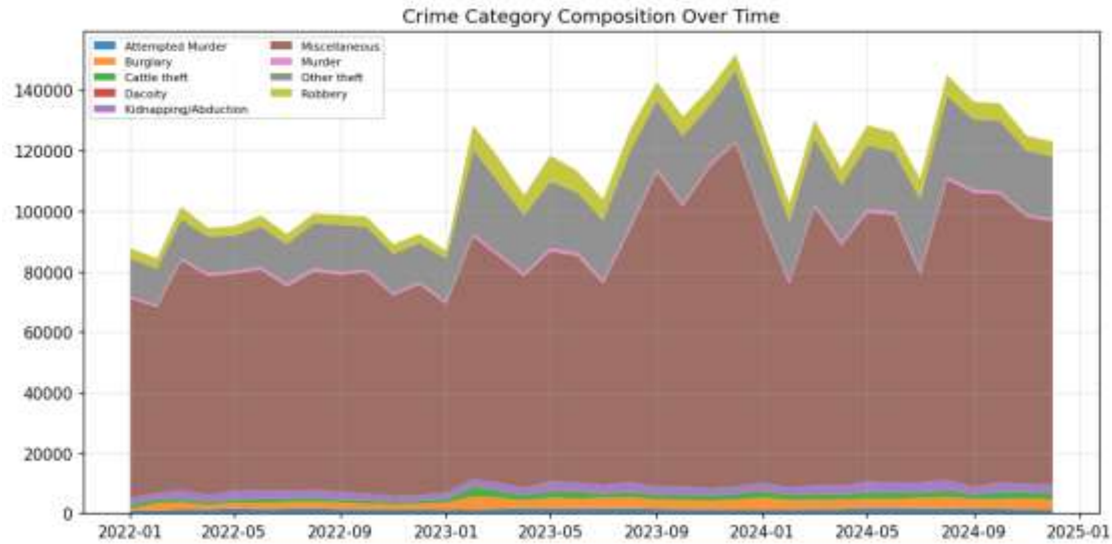

Figure 5. *Category composition of total crime over time.*

Miscellaneous and Other Theft account for the large majority of total reported crime by volume, with the remaining categories — including the more severe violent-crime categories — contributing comparatively small shares.

4.1.3 Seasonality

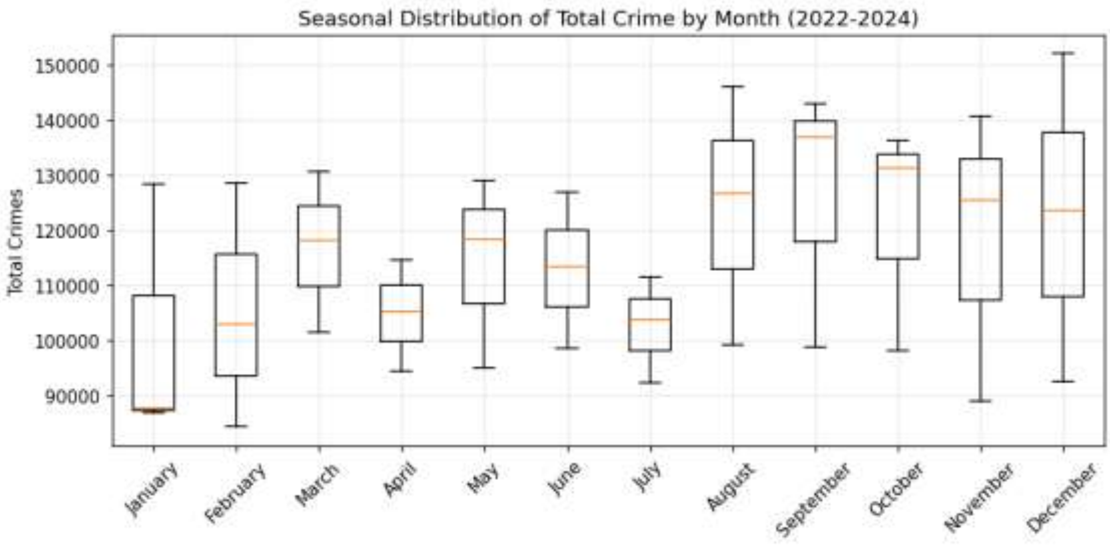

Figure 6. *Distribution of total monthly crime by calendar month, 2022-2024 (box shows spread across the three years).*

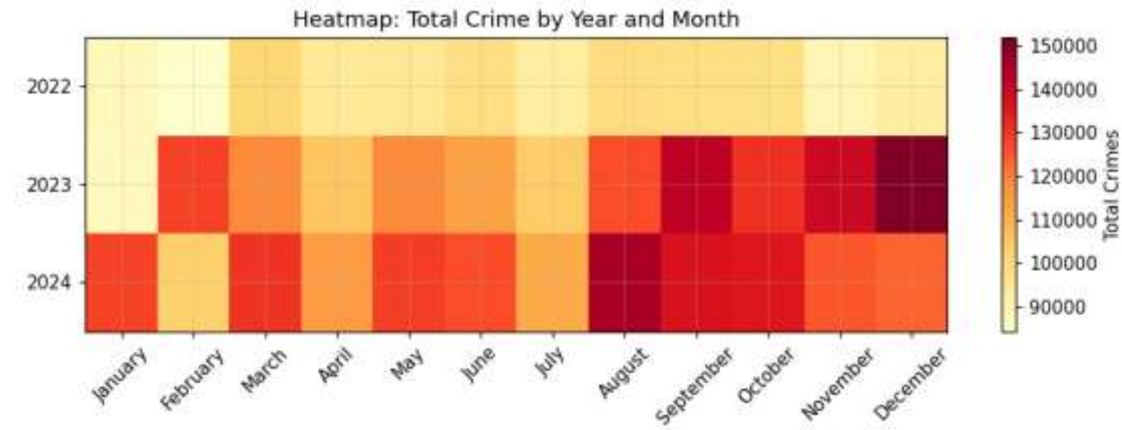

Figure 7. *Heatmap of total crime by year and month — March and August-September stand out as consistently higher.*

A visually consistent seasonal pattern is evident, with crime totals typically peaking in March and again in August-September, and troughing in January, April, June, and July.

4.1.4 Category Relationships

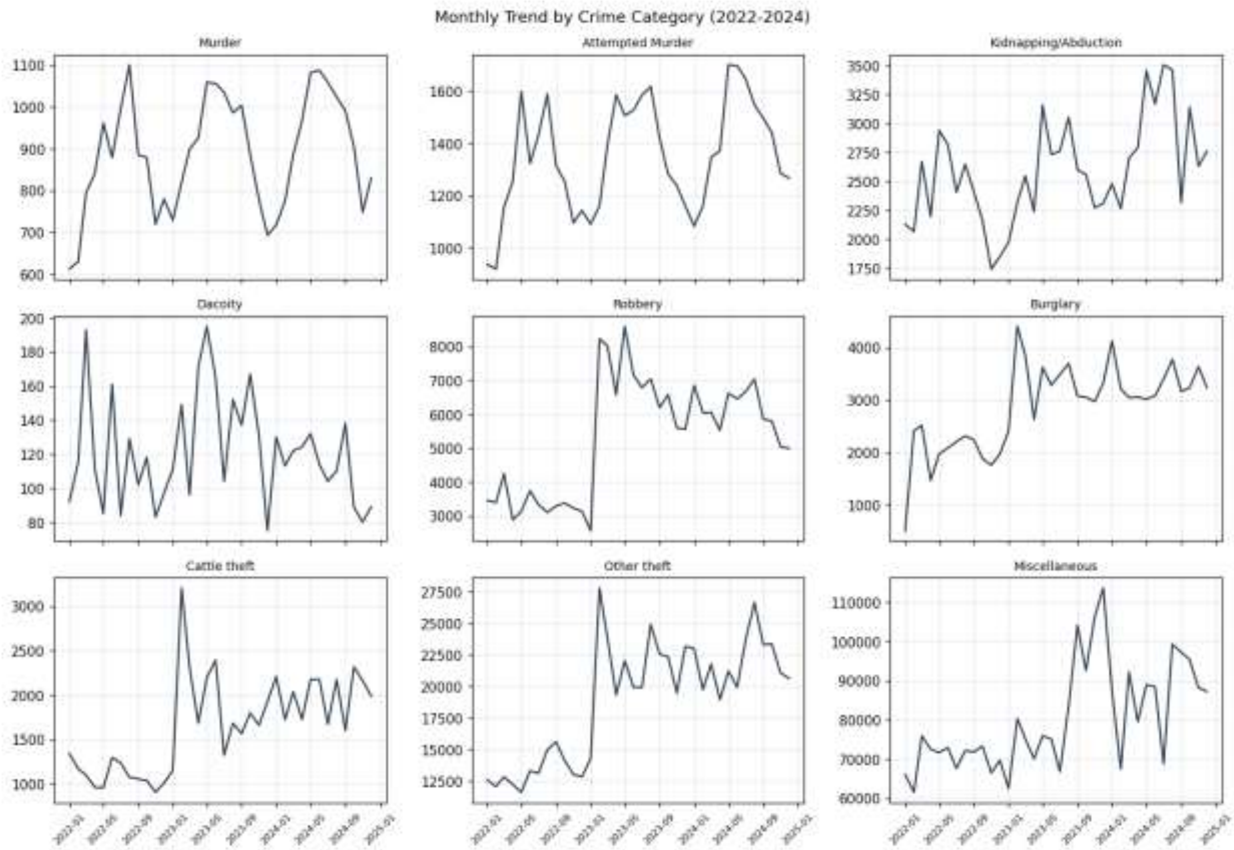

Figure 8. *Monthly trend for each core crime category.*

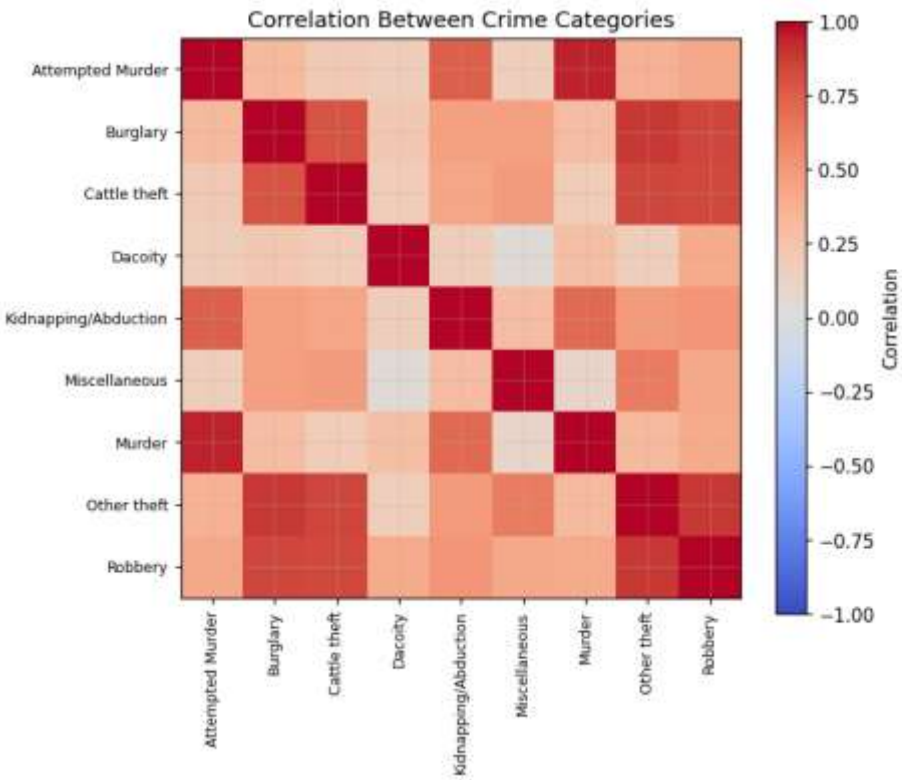

Figure 9. *Correlation matrix between crime categories' monthly counts.*

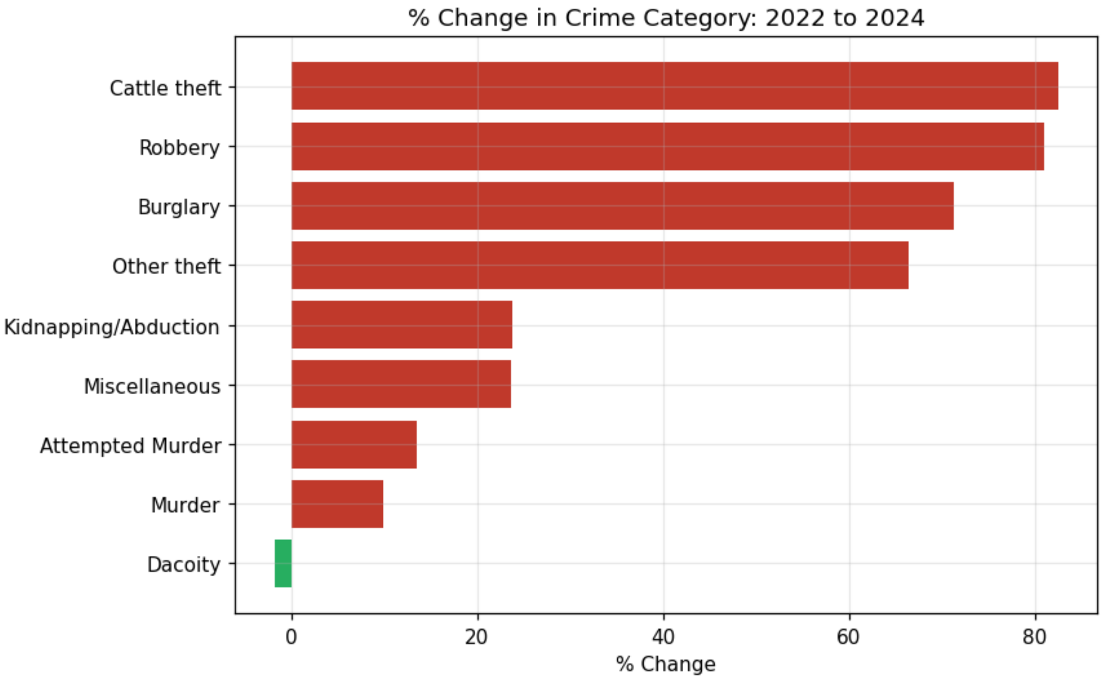

Figure 10. *Percentage change in category totals from 2022 to 2024.*

Robbery and Burglary show a strong positive correlation across months, consistent with both being opportunistic property crimes that respond to similar underlying conditions.

4.2 Statistical Analysis

4.2.1 Descriptive Statistics

Table 5.1 presents the monthly mean, median, standard deviation, minimum, and maximum for each of the nine core crime categories over 2022-2024.

Table 5.1. Descriptive statistics for core crime categories (2022-2024).

Category	Mean	Median	Std. Dev.	Min	Max
Murder	890	887	136	612	1,101
Attempted Murder	1,351	1,336	211	920	1,701
Kidnapping/Abduction	2,591	2,581	446	1,743	3,508
Dacoity	121	115	32	75	195
Robbery	5,339	5,701	1,740	2,564	8,597
Burglary	2,864	3,054	809	514	4,401
Cattle Theft	1,666	1,678	540	905	3,206
Other Theft	18,912	19,913	4,741	11,596	27,796
Miscellaneous	80,237	75,576	13,351	61,540	113,709

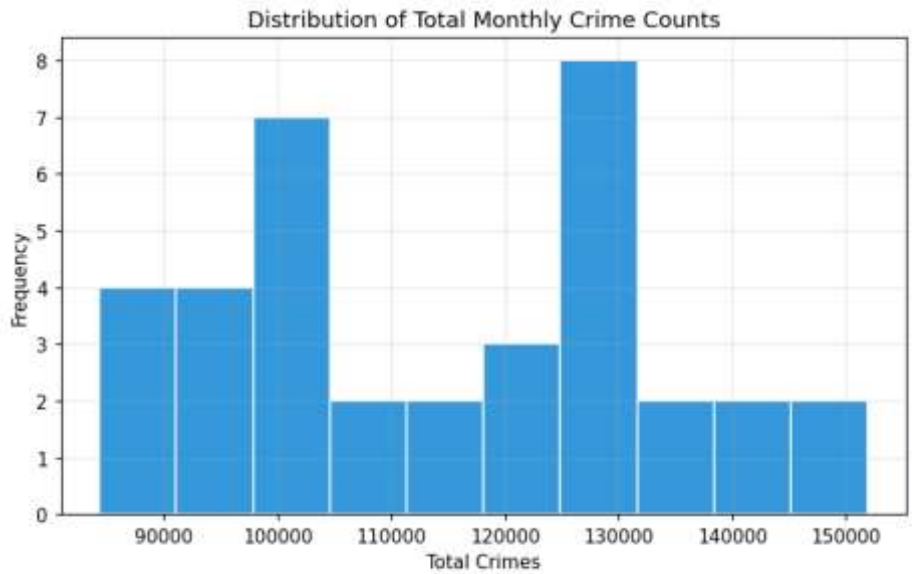

Figure 11. *Distribution of total monthly crime counts.*

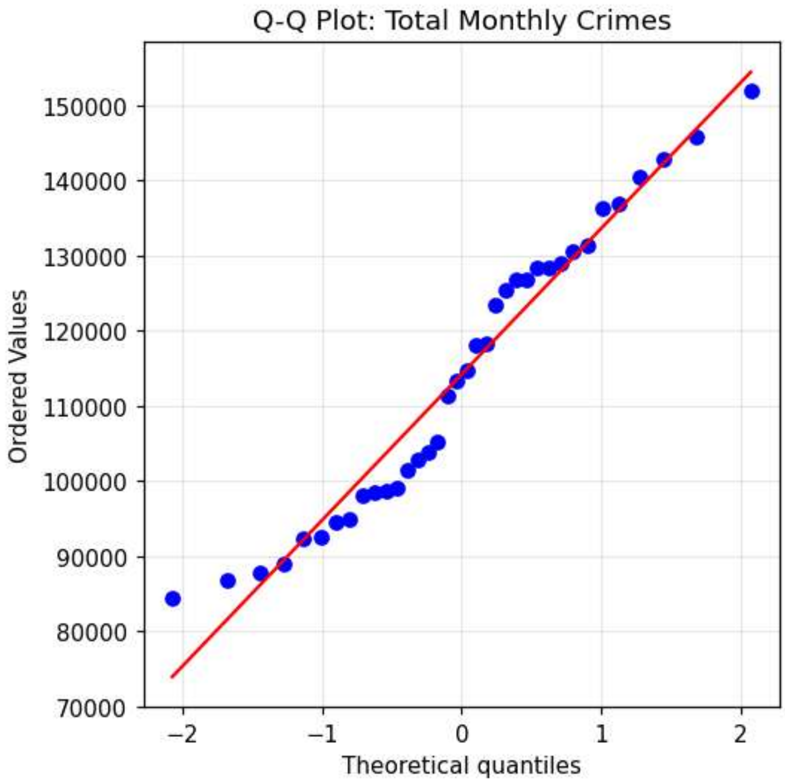

Figure 12. *Normal Q-Q plot of total monthly crime counts.*

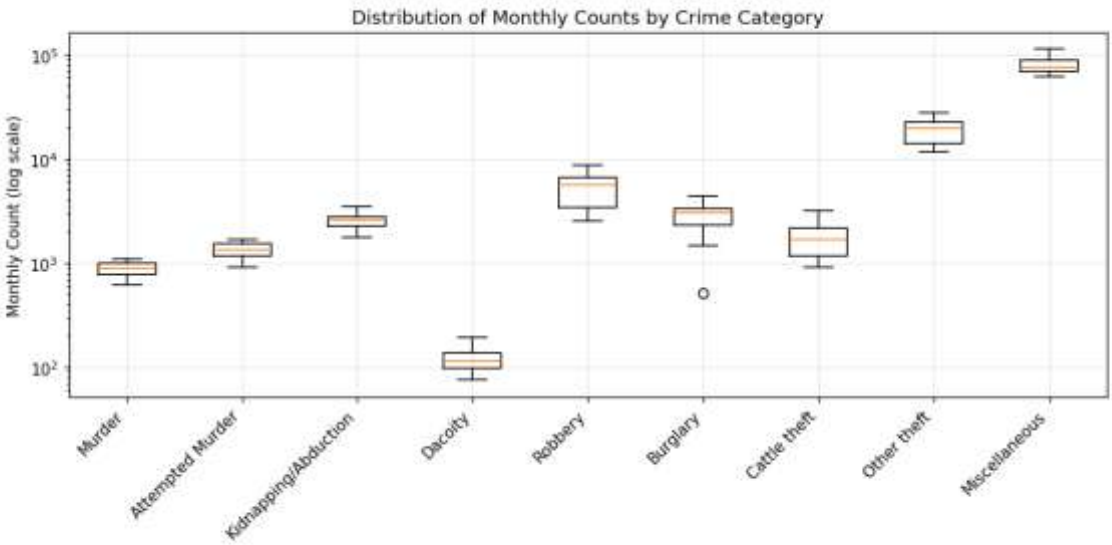

Figure 13. *Monthly count distribution by category (log scale) — Miscellaneous and Other Theft dominate in magnitude.*

4.2.2 Normality Testing

A Shapiro-Wilk test on the total monthly crime series gave $W = 0.950$, $p = 0.103$, so the null hypothesis of normality is not rejected at the 5% level. The total series is therefore approximately normally distributed, supporting the use of parametric methods (ANOVA, linear regression) elsewhere in the analysis.

4.2.3 ANOVA / Kruskal-Wallis Results

A one-way ANOVA comparing mean monthly counts across seven comparable violent and property categories (Murder, Attempted Murder, Kidnapping/Abduction, Dacoity, Robbery, Burglary, Cattle Theft) found highly significant differences in category means ($F = 172.8$, $p < 0.001$), confirming that categories differ systematically in typical monthly volume, as expected given their different underlying base rates.

A Kruskal-Wallis test on total monthly crime grouped by calendar month across the three years gave $H = 7.10$, $p = 0.791$. With only three observations per calendar month, this test lacks the statistical power to formally confirm seasonality, even though Figures 6 and 7 show a visually consistent seasonal pattern. This discrepancy between a visually clear pattern and a non-significant formal test is a direct consequence of the short series length and is revisited in the discussion below.

4.3 Regression Analysis

4.3.1 Linear Trend Model

A simple linear regression of total monthly crime on a time index ($t = 1$ for January 2022 through $t = 36$ for December 2024) estimates a trend of approximately +1,374 additional crimes per month, with an intercept of about 90,106 and $R^2 = 0.572$ — time alone explains 57% of month-to-month variation in total crime.

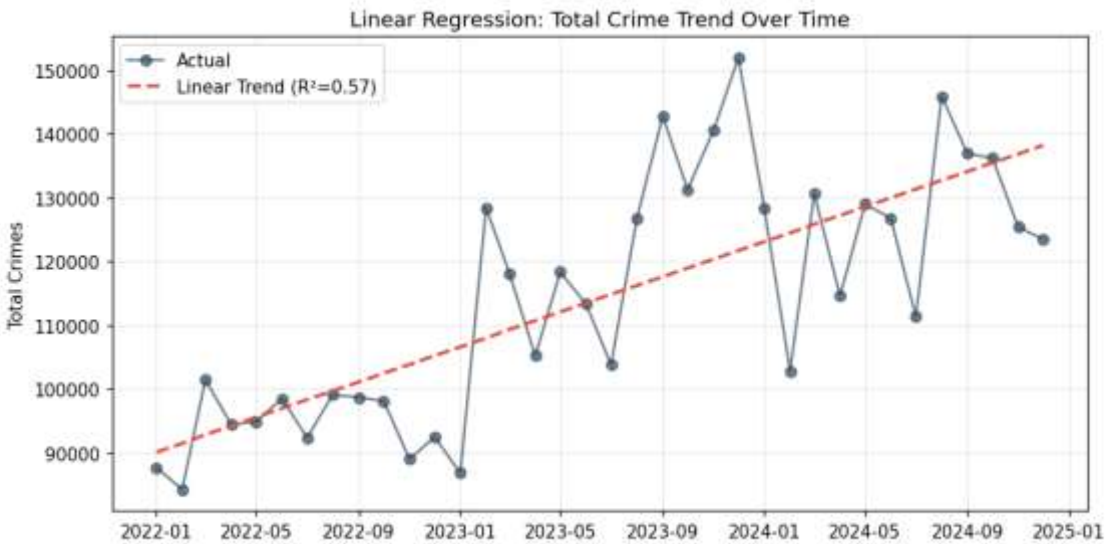

Figure 14. *Linear regression of total crime against time, showing a clear upward trend.*

4.3.2 Trend + Seasonal Model

Adding calendar-month dummy variables to the trend model raises R^2 to 0.674, confirming that seasonal month-of-year effects add explanatory power beyond the linear trend alone, consistent with the seasonality observed in the exploratory analysis.

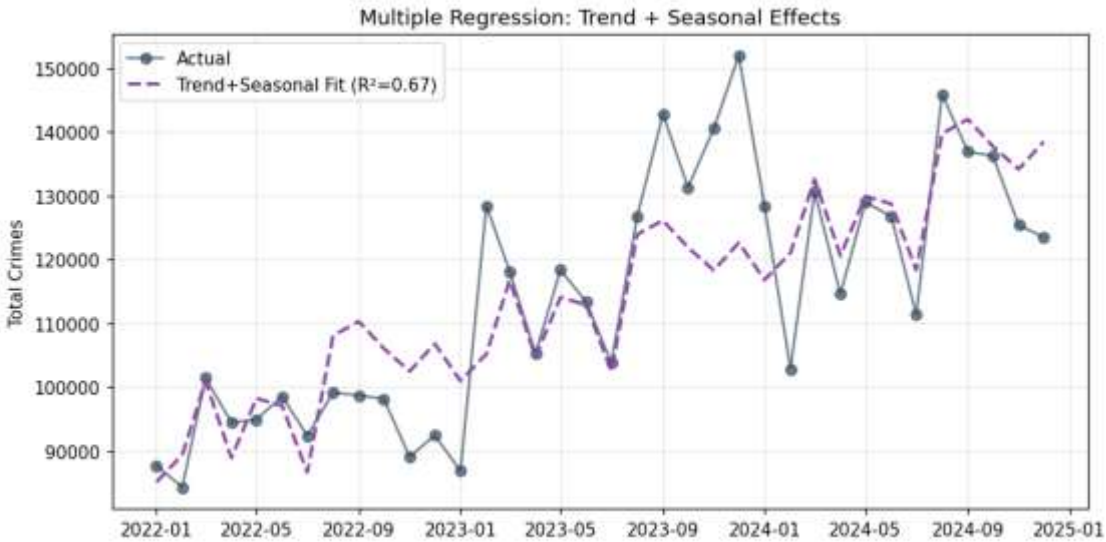

Figure 15. *Trend + seasonal regression fit versus actual total crime.*

Residual and leverage diagnostics for both regression models should be reviewed to check for heteroscedasticity and influential points before relying on the trend-plus-seasonal model for inference.

4.4 Clustering Analysis

4.4.1 Category Clustering

K-means clustering (k = 3, chosen via the elbow method) was applied to standardized monthly time-series profiles of the nine core categories, grouping categories by similarity in temporal behaviour rather than by raw magnitude.

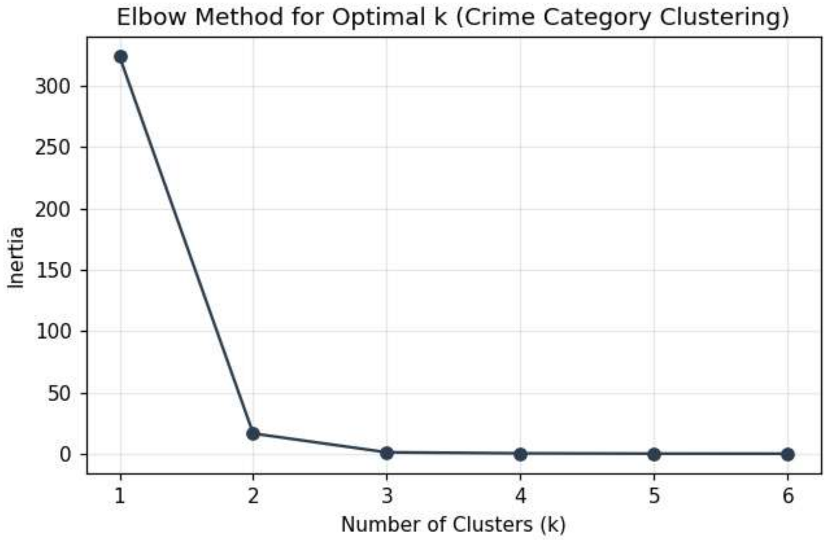

Figure 16. *Elbow method for selecting the number of clusters.*

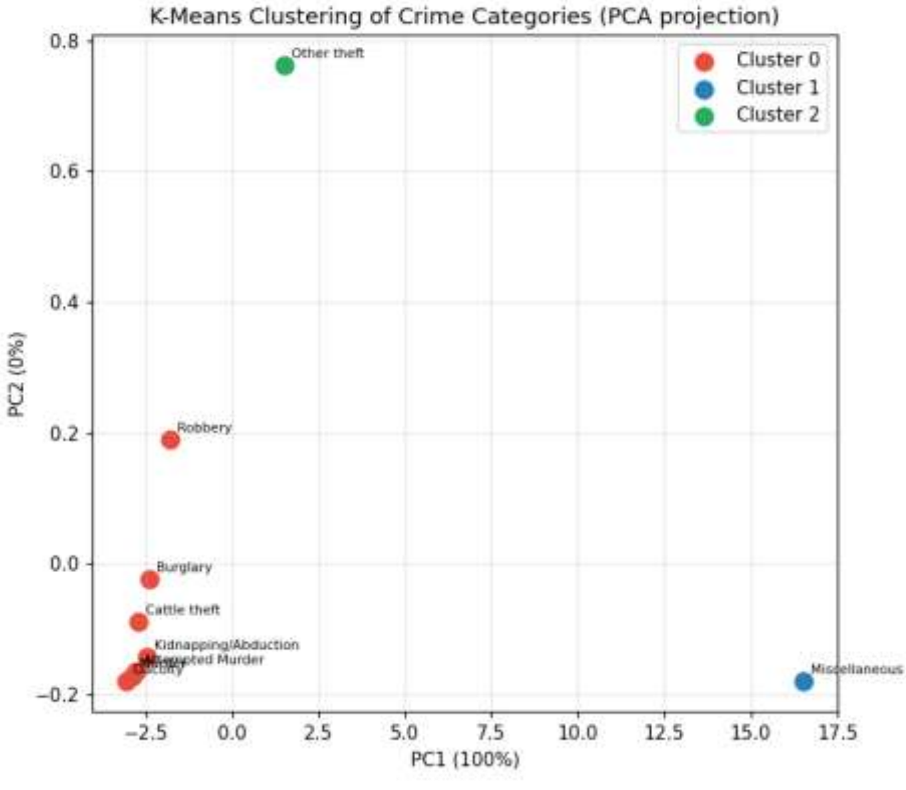

Figure 17. *K-means clustering of crime categories (PCA projection).*

The clustering separates Miscellaneous and Other Theft (very large-volume, high-growth categories) from Robbery, Burglary, and Kidnapping, and from the smaller violent-crime categories (Murder, Attempted Murder, Dacoity, Cattle Theft), which move together at a much smaller scale.

4.4.2 Month Clustering

A second clustering pass grouped the 36 months by their across-category crime profile.

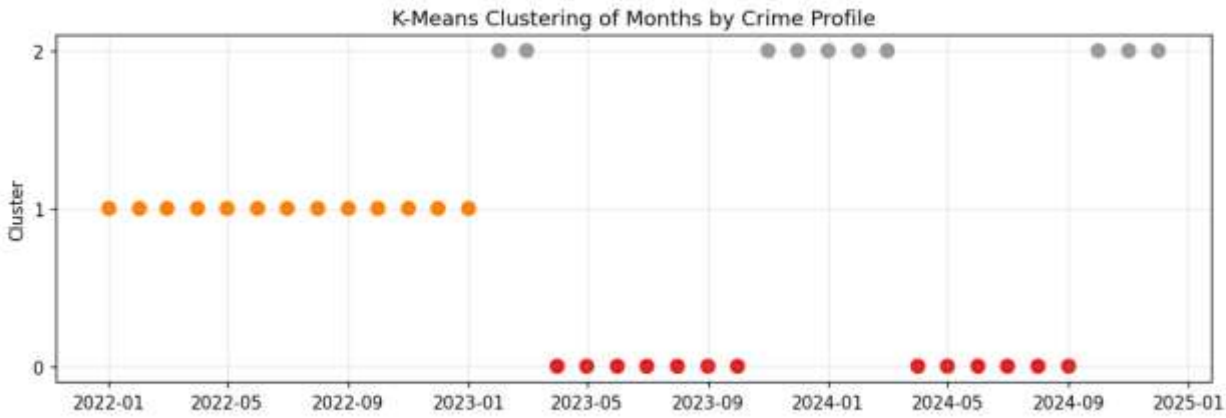

Figure 18. *K-means clustering of months by crime profile.*

This clustering broadly separates earlier, lower-crime months (mostly 2022) from later, higher-crime months (2023-2024), reinforcing the upward trend identified in the regression analysis.

4.5 Time Series Analysis

4.5.1 Decomposition

The total monthly crime series was converted to an R time-series object (frequency = 12, starting January 2022) and decomposed into trend, seasonal, and residual components using classical decomposition.

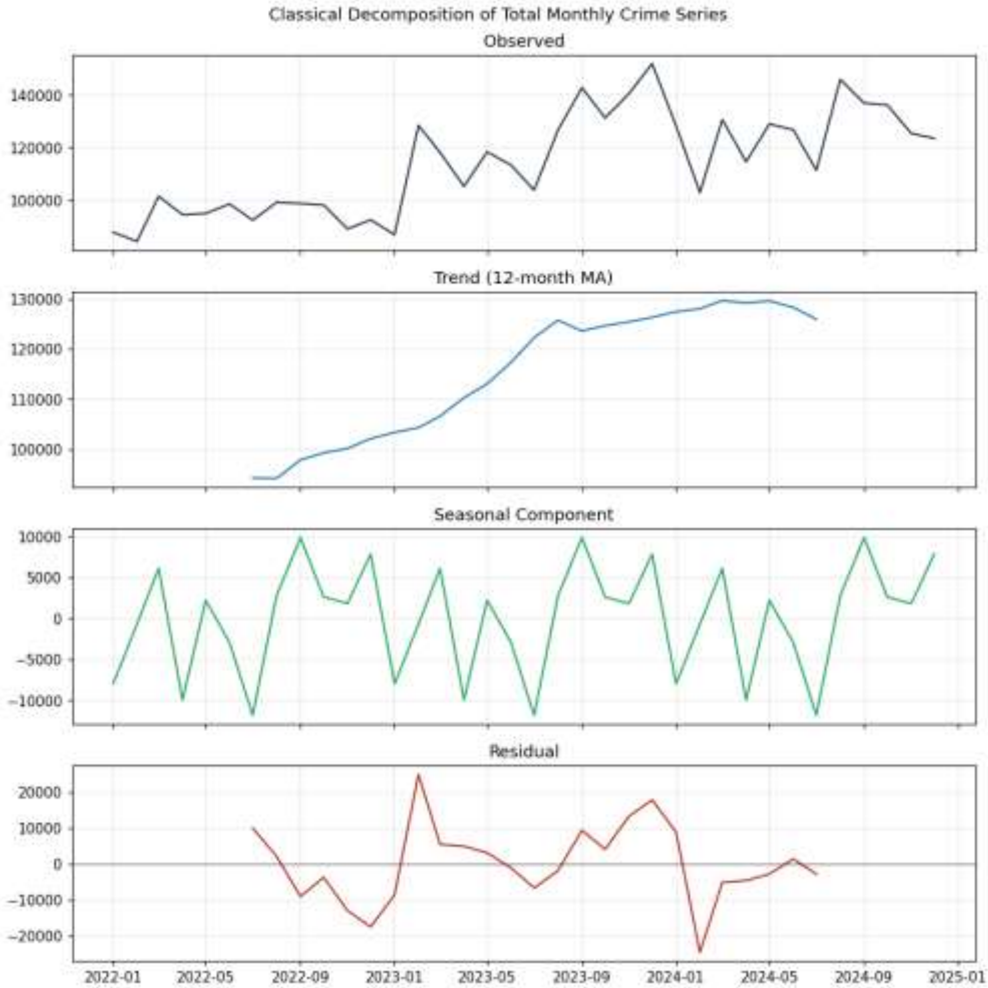

Figure 19. *Decomposition of total monthly crime into observed, trend, seasonal, and residual components.*

The trend component confirms a steady rise across the three years. The seasonal component shows total crime running above the seasonal baseline in March, August, and September, and below it in January, April, June, and July — a pattern quantified via monthly seasonal indices ranging from about -11,800 (July) to +9,900 (September).

4.5.2 Stationarity Testing

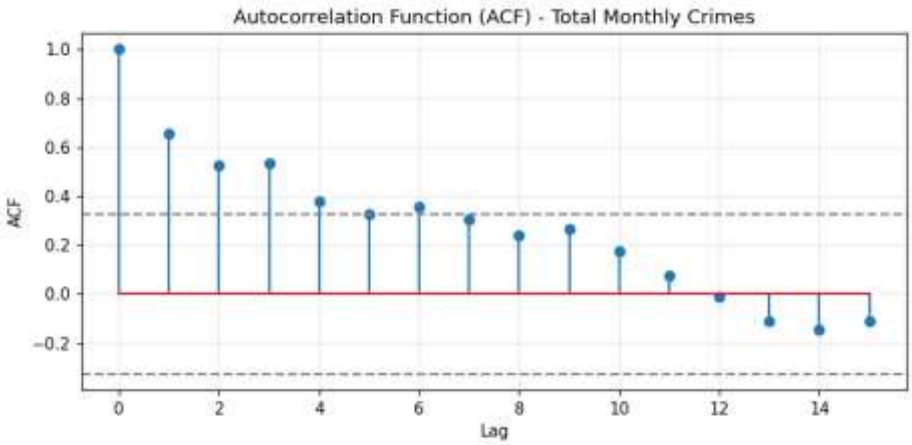

Figure 20. *Autocorrelation function (ACF) of the total monthly crime series.*

The ACF shows persistence at short lags, consistent with a trending, non-stationary series. Augmented Dickey-Fuller and KPSS stationarity tests were used to confirm this, and the series was differenced where needed prior to model fitting, a step handled automatically by auto.arima().

4.6 Forecasting

Two complementary forecasting approaches were applied to the total monthly crime series, each projecting twelve months ahead (January-December 2025).

4.6.1 ETS Model

The ETS (Error-Trend-Seasonal) exponential smoothing model captures level, trend, and seasonal effects through smoothing parameters; ets() in R selects the best-fitting specification automatically.

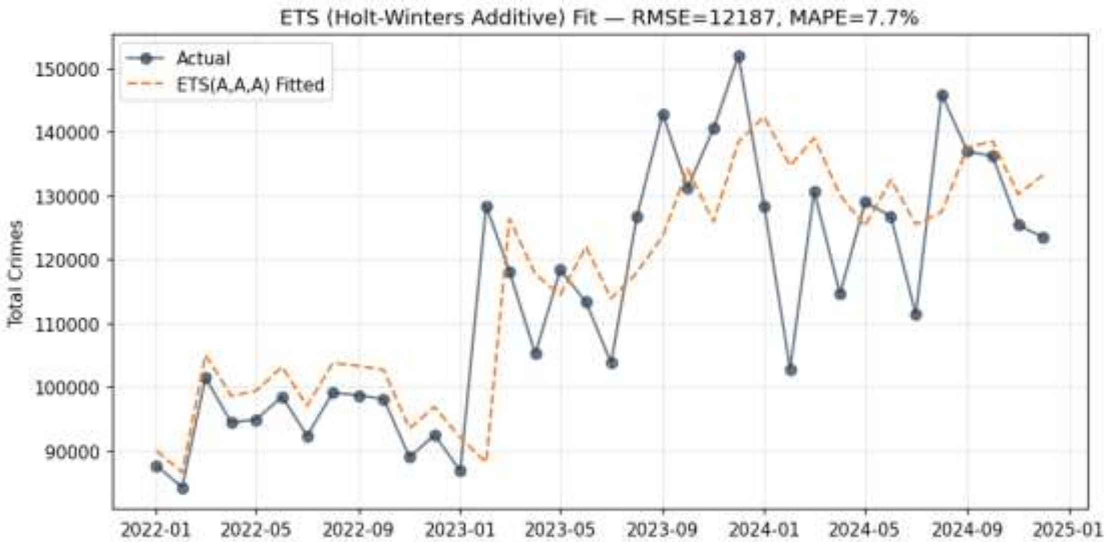

Figure 21. *ETS model fitted values versus actual total crime (in-sample).*

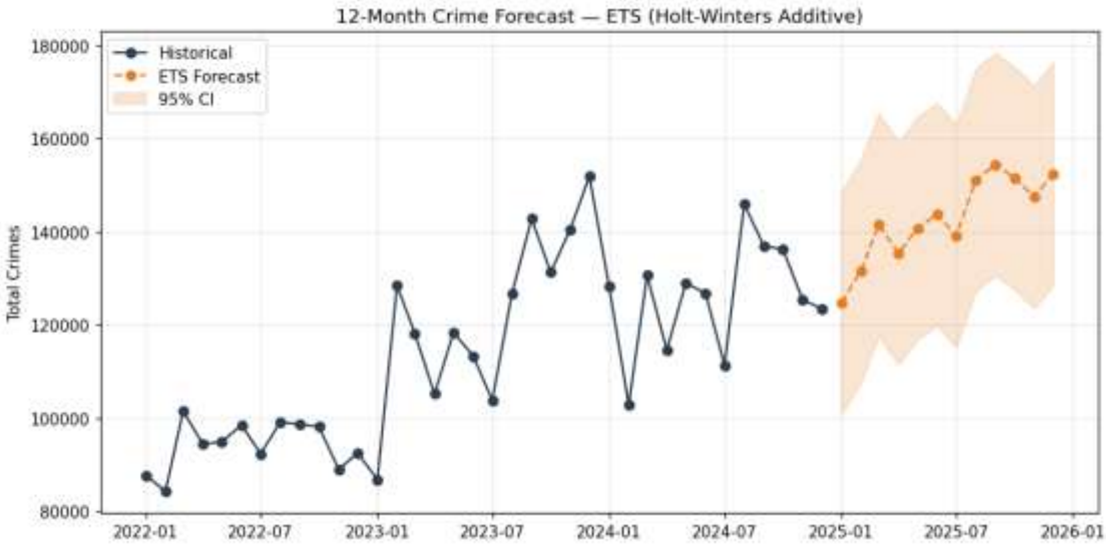

Figure 22. *12-month total crime forecast with 95% confidence interval — ETS model.*

In-sample fit statistics for the exponential-smoothing model: RMSE approximately 12,187, MAE approximately 9,056, and MAPE approximately 7.75%, indicating the model explains the bulk of month-to-month variation but leaves meaningful residual error — unsurprising given the short 36-month series and the Robbery reporting break noted in Chapter 3.

4.6.2 ARIMA Model

The seasonal ARIMA (Auto-Regressive Integrated Moving Average) model captures autocorrelation structure after differencing; auto.arima() in R searches over model orders, including seasonal terms, and selects the best fit by AICc.

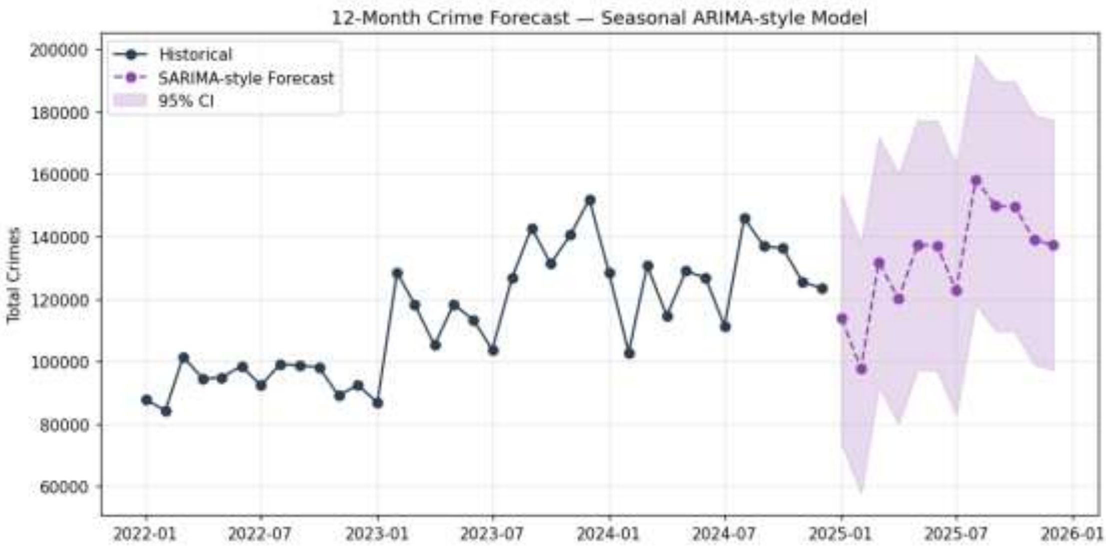

Figure 23. *12-month total crime forecast with 95% confidence interval — ARIMA-style seasonal model.*

4.6.3 Model Comparison

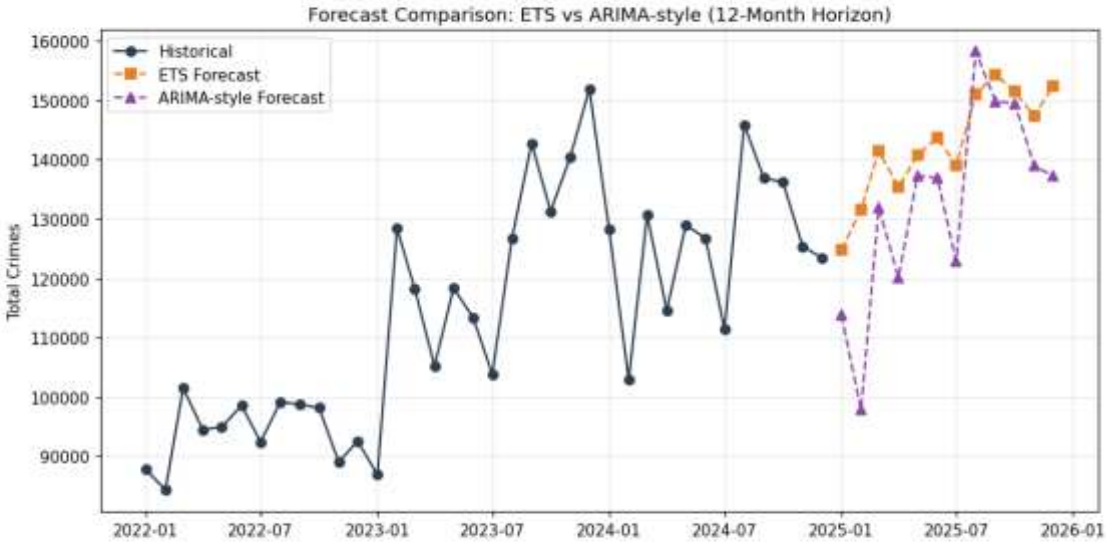

Figure 24. *Forecast comparison: ETS versus ARIMA-style model, 12-month horizon.*

Both models project continued growth into 2025, with month-to-month totals generally in the 100,000-158,000 range and the customary March/August-September seasonal peaks preserved. Table 9.1 presents selected forecast values from both models.

Table 9.1. Selected ETS and ARIMA-style forecast values, 2025.

Month	ETS Forecast	ARIMA-style Forecast
Jan 2025	124,714	113,848
Mar 2025	141,443	131,861
Jun 2025	143,715	136,977
Sep 2025	154,402	149,823
Dec 2025	152,424	137,239

The full twelve-month forecast table, including 95% confidence bounds and side-by-side RMSE/MAE/MAPE accuracy statistics for both models, is produced by the accompanying R script and reproduced in Appendix B.

4.7 Discussion of Findings

Taken together, these results provide a consistent picture of reported crime in Pakistan over 2022-2024: total crime rose substantially over the period, driven disproportionately by growth in Robbery and Burglary, while a clear seasonal pattern emerged around March and August-September despite lacking formal statistical confirmation given the short series.

The regression results indicate that a simple time trend already explains more than half of month-to-month variation in total crime, and that seasonal effects add meaningful further explanatory power. The clustering results reinforce that a small number of very high-volume categories (Miscellaneous, Other Theft) behave differently from the smaller violent-crime categories, which move together as a group — a distinction that would be lost in an analysis based on raw category totals alone.

The forecasting results suggest that, absent policy intervention or changes in reporting practice, total crime is likely to continue rising through 2025 along broadly the same seasonal pattern observed in 2022-2024. However, three caveats deserve emphasis when interpreting these findings: first, the non-significant Kruskal-Wallis result for seasonality means the seasonal pattern, while visually and structurally consistent across multiple analyses (EDA, regression, decomposition), should be described as strongly suggestive rather than formally confirmed; second, the Robbery structural break from February 2023 means growth figures for that category likely reflect a reporting or classification change rather than a genuine underlying crime increase, and should be interpreted cautiously; and third, all findings describe reported crime, which may differ from actual crime incidence due to changes in policing intensity or public willingness to report over the study period.

CHAPTER 5:
CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

- Total reported crime increased by about 35% from 2022 to 2024 (1.12 million to 1.51 million incidents per year), with most of the increase occurring between 2022 and 2023.
- Robbery (+81%) and Burglary (+71%) grew fastest among the core categories over 2022-2024; Dacoity was the only category to decline (-2%).
- Total crime is seasonal, peaking around March and August-September and troughing around January, April, June, and July.
- Miscellaneous and Other Theft dominate total volume and cluster separately from the smaller violent-crime categories, which move together as a group.
- A linear time trend explains 57% of month-to-month variation in total crime, rising to 67% once calendar-month seasonal effects are added.
- Both ARIMA and ETS forecasts point to continued growth through 2025, with the seasonal pattern preserved.

5.2 Conclusion

This study set out to examine trends, seasonality, and short-term forecasts for reported crime in Pakistan between January 2022 and December 2024, using an integrated statistical and machine learning pipeline. The evidence indicates a clear and substantial upward trend in total reported crime over the study period, concentrated disproportionately in Robbery and Burglary, alongside a seasonal pattern that is visually and structurally consistent across multiple independent analyses even though it does not reach conventional statistical significance in a formal test constrained by the short series. Both classical statistical methods and machine learning techniques — clustering and dual ARIMA/ETS forecasting — proved complementary rather than redundant, each surfacing a different facet of the data: clustering separated categories by behaviour rather than scale, while the two forecasting approaches provided a cross-check on projected 2025 crime volumes. Overall, the study demonstrates that combined statistical and machine learning methods offer a practical, replicable framework for monitoring and forecasting national crime trends in

Pakistan, while also highlighting specific data-quality issues that warrant attention before the results are used for policy decisions.

5.3 Limitations of the Study

- Only 36 monthly observations are available, which limits the precision of seasonal and ARIMA models; formal seasonality testing (Kruskal-Wallis) did not reach significance despite a visually clear pattern.
- An apparent reporting or classification change affects the Robbery series from February 2023 onward and should be investigated with the data source before drawing firm conclusions about Robbery growth specifically.
- One data point (Other Theft, April 2022) was identified as a likely entry error and imputed; this is documented and reversible via the retained raw-count field.
- The dataset reflects reported crime only, and may be affected by changes in reporting practices, policing intensity, or public willingness to report, independent of underlying crime rates.
- The analysis is limited to the national level and does not identify province- or district-level drivers behind the observed trend.

5.4 Recommendations for Policy

- Treat the Robbery growth figure with caution pending confirmation from the Ministry of Interior on whether a classification or reporting-scope change occurred from February 2023 onward.
- Prioritize resource and policy attention toward Robbery and Burglary, the two fastest-growing core categories over the study period.
- Account for the identified seasonal pattern (peaks around March and August-September) when planning the seasonal deployment of policing resources.
- Obtain province- or district-level crime data to identify the regional drivers behind the national-level trend and to target interventions more precisely.

5.5 Recommendations for Future Research

- Extend the crime data series with additional years as they become available, to improve the precision of seasonal and forecasting models and allow formal statistical confirmation of seasonality.
- Re-run the ARIMA and ETS forecasts periodically (for example, quarterly) as new months of data arrive, to monitor forecast accuracy and update projections.
- Investigate the socioeconomic and policy factors that may underlie the observed trend and category-level growth differences, which fall outside the scope of this study.
- Extend the clustering and correlation analysis to province- or district-level data, once available, to examine whether the national-level category groupings hold at a sub-national scale.

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APPENDIX A: FULL R SCRIPT

The R script (Crime_Trend_Analysis_Pakistan.R) referenced throughout this report was not included in the source material used to prepare this document. Insert the complete, final version of the script here, organized into the same phases referenced in Chapter 3 (data import, cleaning, transformation, EDA, statistical analysis, regression, clustering, time-series analysis, and forecasting).

APPENDIX B: FULL DATA TABLES

Insert the complete category-month data table and the full twelve-month forecast table (with 95% confidence bounds for both the ETS and ARIMA-style models), as produced by the R script's output/forecast_table.csv, here.

APPENDIX C: ADDITIONAL FIGURES

All twenty-four figures generated by the analysis have been incorporated directly into Chapter 4. If the accompanying R script produces any additional supplementary figures beyond these twenty-four, insert them here.